

Algebraic Curves

Chapter 8.4. Derivations and Differentials

Let $\mathfrak{R}$ be a ring containing $\mathbb{k}$ and $\mathfrak{M}$ be a $\mathfrak{R}$ -module. A ‘derivation’ of $\mathfrak{R}$ into $\mathfrak{M}$ over $\mathbb{k}$ is a $\mathbb{k}$ -linear map $D: \mathfrak{R} \rightarrow \mathfrak{M}$ satisfying the product rule,

$$D[xy] = x D[y] + y D[x]$$

Or,

$$D[F(x_1, \dots, x_n)] = \sum_{i=1}^n \frac{\partial F}{\partial x_i} [x_1, \dots, x_n] D[x_i]$$

LEMMA 2.

If $\mathfrak{R}$ is a domain s.t. $\text{Frac } \mathfrak{R} = \mathbb{K}$ and $\mathfrak{M}$ is a vector space over $\mathbb{K}$, then derivations $D: \mathfrak{R} \rightarrow \mathfrak{M}$ extend uniquely to a derivation $D: \mathbb{K} \rightarrow \mathfrak{M}$.

PROOF.

Let $z \in \mathbb{K}$ and write $z = x/y$ for $x, y \in \mathfrak{R}$. Hence $x = yz$. Thus,

$$D x = y D z + z D y$$

And so,

$$D z = \frac{D x - z D y}{y}$$

The formula gives the uniqueness. Defining the extension this way gives a derivation $\mathbb{K} \rightarrow \mathfrak{M}$. This is the quotient rule. ■

Let $\mathfrak{F}$ be the free $\mathfrak{R}$ -module on the set $\{[x] \mid x \in \mathfrak{R}\}$ where the symbol is so defined to have its relations encompassed by the submodule $\mathfrak{N}$ generated by,

1. $[x + y] - [x] - [y]$ for $x, y \in \mathfrak{R}$.
2. $[\lambda x] - \lambda[x]$ for $x \in \mathfrak{R}, \lambda \in \mathbb{k}$.

3. $[xy] - x[y] - y[x]$ for $x, y \in \mathfrak{A}$.

Then let $\Omega_{\mathbb{k}}[\mathfrak{A}] = \mathfrak{F}/\mathfrak{M}$. Denote dx the image of $[x]$ in $\mathfrak{F}/\mathfrak{M}$ and $d: \mathfrak{A} \rightarrow \Omega_{\mathbb{k}}[\mathfrak{A}]$ the map that takes x to dx . $\Omega_{\mathbb{k}}[\mathfrak{A}]$ is an $\mathfrak{A}$ -module known as the ‘module of differentials’ of $\mathfrak{A}$ over $\mathbb{k}$, and $d: \mathfrak{A} \rightarrow \Omega_{\mathbb{k}}[\mathfrak{A}]$ a ‘derivation’.

LEMMA 3.

For any $\mathfrak{A}$ -module $\mathfrak{M}$ and any derivation $D: \mathfrak{A} \rightarrow \mathfrak{M}$, there exists a unique $\mathfrak{A}$ -linear homomorphism $\varphi: \Omega_{\mathbb{k}}[\mathfrak{A}] \rightarrow \mathfrak{M}$ s.t. $Dx = \varphi[dx]$ for all $x \in \mathfrak{A}$.

PROOF.

Define,

$$\varphi': \mathfrak{F} \rightarrow \mathfrak{M}, \quad \sum_i x_i [y_i] \mapsto \sum_i x_i D y_i$$

$\mathfrak{M}$ is the submodule of $\mathfrak{F}$ whose relations are exactly that which vanishes for the derivation D , so that $\varphi'[\mathfrak{M}] = 0$. Therefore we induce the well-defined map on the quotient $\varphi: \Omega_{\mathbb{k}}[\mathfrak{A}] \rightarrow \mathfrak{M}$.

■

Namely observe that over the polynomial ring $G \in \mathbb{k}[X_1, \dots, X_n]$ and $x_1, \dots, x_n \in \mathfrak{A}$ we exactly have,

$$dG[x_1, \dots, x_n] = \sum_{i=1}^n \frac{\partial G}{\partial X_i} [x_1, \dots, x_n] dx_i$$

In particular if $\mathfrak{A} = \mathbb{k}[x_1, \dots, x_n]$, then $\Omega_{\mathbb{k}}[\mathfrak{A}] = \mathfrak{A}dx_1 + \dots + \mathfrak{A}dx_n$. LEMMA 2. also says that this extends to the quotient field $\text{Frac } \mathfrak{A} = \mathbb{K}$ where we may define for $z = x/y$, $dz = y^{-1}dx - y^{-1}zdy$. Hence we show that if $\mathbb{K} = \mathbb{k}(x_1, \dots, x_n)$, then $\Omega_{\mathbb{k}}[\mathbb{K}] = \mathbb{K}dx_1 + \dots + \mathbb{K}dx_n$.

PROPOSITION 6.

Let $\mathbb{K}$ be an algebraic function field in one variable over $\mathbb{k}$. Then $\Omega_{\mathbb{k}}[\mathfrak{A}]$ is a one-dimensional vector space over $\mathbb{K}$. Moreover (when $\text{char } \mathbb{k} = 0$) if $x \in \mathbb{K} \setminus \mathbb{k}$, then dx is its basis.

PROOF.

Let $F \in \mathbb{k}[X, Y]$ be the affine plane curve whose function field is $\mathbb{K}$. Denote $\mathfrak{A} = \mathbb{k}[X, Y]/\langle F \rangle = \mathbb{k}[x, y]$ and $\mathbb{K} = \mathbb{k}(x, y)$. W.l.o.g. by switching variables let $\partial F / \partial Y \neq 0$ by irreducibility of F . Namely this means F does not divide $\partial F / \partial Y$, i.e. $\partial F / \partial Y [x, y] \neq 0$. We have $\Omega_{\mathbb{k}}[\mathbb{K}] = \mathbb{K}dx + \mathbb{K}dy$. However,

$$0 = dF[x, y] = \frac{\partial F}{\partial X} [x, y] dx + \frac{\partial F}{\partial Y} [x, y] dy$$

So that $dy = u dx$ for $u \in \mathbb{K}$ the prefactor by rearranging. Thus dx is sufficient to generate $\Omega_{\mathbb{k}}[\mathbb{K}]$, so its dimension is less-than-or-equal-to one.

To see that $\Omega_{\mathbb{K}}[\mathbb{K}]$ is nonempty it suffices to find a nonzero derivation $D: \mathfrak{A} \rightarrow \mathfrak{M}$ for some vector space $\mathfrak{M}$ over $\mathbb{K}$. Consider $\mathfrak{M} = \mathbb{K}$ and for any $G \in \mathbb{k}[X, Y]$ denote $\bar{G}$ its image in $\mathfrak{A}$. Define D by the map,

$$D \bar{G} = \frac{\partial G}{\partial X}[x, y] - u \frac{\partial G}{\partial Y}[x, y]$$

Then indeed D is well-defined and $Dx = 1$, so is nontrivial. ■

Thus for any $f, t \in \mathbb{K}$ where $t \notin \mathbb{k}$, there is a unique element $v \in \mathbb{K}$ s.t. $df = vdt$. We denote this v by df/dt , and call it the ‘derivative’ of f w.r.t t .

PROPOSITION 7.

With $\mathbb{K}$ as in the previous proposition let $\mathcal{O}$ be its d.v.r. and t its uniformizing parameter. If $f \in \mathcal{O}$, then $df/dt \in \mathcal{O}$.

PROOF.

Observe $\mathcal{O} = \mathcal{O}_{\mathbf{p}}[F]$ and w.l.o.g. let $\mathbf{p} = \mathbf{0}$ simple on $\{F = 0\}$. Fix t and for $z \in \mathbb{K}$ denote $\dot{z} := dz/dt$. Choose N large s.t. $\text{ord}_{\mathbf{p}}[\dot{x}], \text{ord}_{\mathbf{p}}[\dot{y}] \geq -N$. Therefore if $f \in \mathfrak{A} = \mathbb{k}[x, y]$, then $\text{ord}_{\mathbf{p}}[f] \geq -N$ as,

$$\dot{f} = \frac{\partial f}{\partial X}[x, y]\dot{x} + \frac{\partial f}{\partial Y}[x, y]\dot{y}$$

And use of the reverse triangle inequality. Hence let $f \in \mathcal{O}$ and write $f = g/h$. Therefore $\dot{f} = h^{-2}(h\dot{g} - g\dot{h})$. Again using the reverse triangle inequality and regularity of h at $\mathbf{p}$ we have $\text{ord}_{\mathbf{p}}[\dot{f}] \geq -N$. Consequently we are able to write in generality,

$$f = \sum_{i < N} \lambda_i t^i + t^N g$$

This being the decomposition of $f \in \mathcal{O}$ where $\lambda_i \in \mathbb{k}, g \in \mathcal{O}$. Hence,

$$\dot{f} = \sum_{i < N} i \lambda_i t^{i-1} + N t^{N-1} g + t^N \dot{g}$$

Thus as $\text{ord}_{\mathbf{p}}[\dot{g}] \geq -N$ and each other term is in $\mathcal{O}$, so too is $\dot{f}$. ■

Question 8.22

PROPOSITION.

Let $\mathbb{K}$ be an algebraic function field of n variables over $\mathbb{k}$. Then $\Omega_{\mathbb{k}}[\mathbb{K}]$ is an n -dimensional vector space over $\mathbb{k}$.

PROOF.

Let $F \in \mathbb{k}[X_0, X_1, \dots, X_n]$ be the affine plane curve birationally equivalent to our function field $\mathbb{K}$ in that $\mathbb{K} = \mathbb{k}(F)$. Hence,

$$\mathfrak{R} = \mathbb{k}[X_0, X_1, \dots, X_n]/\langle F \rangle =: \mathbb{k}[x_0, x_1, \dots, x_n], \quad \mathbb{K} = \mathbb{k}(x_0, x_1, \dots, x_n)$$

W.l.o.g. by relabeling assume $\partial F / \partial X_0[x_0, x_1, \dots, x_n] \neq 0$ by irreducibility of F . We know that $\Omega_{\mathbb{k}}[\mathbb{K}]$ is generated by $dx_0, dx_1, \dots, dx_n$ over $\mathbb{K}$. By applying the differential to the equation $F[x_0, x_1, \dots, x_n] = 0$ we obtain,

$$0 = dF[x_0, x_1, \dots, x_n] = \frac{\partial F}{\partial X_0}[x_0, x_1, \dots, x_n]dx_0 + \sum_{i=1}^n \frac{\partial F}{\partial X_i}[x_0, x_1, \dots, x_n]dx_i$$

Rearranging yields,

$$dx_0 = -\frac{1}{\partial F / \partial X_0[x_0, x_1, \dots, x_n]} \sum_{i=1}^n \frac{\partial F}{\partial X_i}[x_0, x_1, \dots, x_n]dx_i$$

This shows that dx_0 is a linear combination of all other dx_i . Hence $\dim \Omega_{\mathbb{k}}[\mathbb{K}] \leq n$.

For the other direction assume for contradiction there was an equation,

$$\sum_i f_i dx_i = 0$$

Define,

$$D_j: \mathfrak{R} \rightarrow \Omega_{\mathbb{k}}[\mathbb{K}], \quad G \mapsto \frac{\partial G}{\partial x_j}$$

That D_j is a derivation follows from Leibnitz's Rule. From LEMMA 2. this extends into a derivation $D_j: \mathbb{K} \rightarrow \Omega_{\mathbb{k}}[\mathbb{K}]$. By LEMMA 3. there exists an $\mathfrak{R}$ -linear homomorphism $\varphi: \Omega_{\mathbb{k}}[\mathbb{K}] \rightarrow \Omega_{\mathbb{k}}[\mathbb{K}]$ s.t. $D_j[x] = \varphi[dx]$. Applying this homomorphism to the initial equation,

$$0 = \varphi[0] = \varphi\left[\sum_i f_i dx_i\right] = \sum_i f_i \varphi[dx_i] = \sum_i f_i D_j[x_i] = f_j$$

This is true for all j . Hence each $f_j = 0$, contradicting linear independence. ■

Question 8.23

PROPOSITION.

Let $\mathcal{O} = \mathcal{O}_{\mathbf{p}}[F]$ and t its uniformizing parameter. Let $\varphi: \mathbb{K} \rightarrow \mathbb{k}((t))$ be the power-series expansion homomorphism. Then for any $f \in \mathcal{O}$, φ maps the derivative of f to its formal derivative in $\mathbb{k}((t))$.

PROOF.

Recall that we have the representation,

$$\varphi[f] = \sum_{i \in \mathbb{Z}} \lambda_i t^i$$

Applying the formal derivative yields,

$$D\varphi[f] = D \left[\sum_{i \in \mathbb{Z}} \lambda_i t^i \right] = \sum_{i \in \mathbb{Z}} i \lambda_i t^{i-1}$$

And this is a derivation. By the universal property there is a $\mathbb{K}$ -linear homomorphism $\psi: \Omega_{\mathbb{K}}[\mathbb{K}] \rightarrow \mathbb{k}((t))$ s.t. $D[x] = \psi[dx]$. Applied here this precisely states,

$$D\varphi[f] = \psi[df]$$

So indeed φ maps df to its formal derivative.

Namely observe that given $f \in \mathcal{O}$, $\text{ord}_{\mathbf{p}}[f] \geq 0$. From the power-series expansion this is equivalent to,

$$0 \leq \text{ord}_{\mathbf{p}}[f] = \text{ord}_{\mathbf{p}} \left[\sum_{i \in \mathbb{Z}} \lambda_i t^i \right] = \min_{i \in \mathbb{Z}} \{ \lambda_i t^i \}$$

That is, 0 is the least possible nonzero exponent in the expansion of f , so our range of indices is $i \geq 0$. Therefore for the formal derivative,

$$\text{ord}_{\mathbf{p}}[\dot{f}] = \text{ord}_{\mathbf{p}} \left[\sum_{i \geq 0} (i+1) \lambda_{i+1} t^i \right] = \min_{i \geq 0} \{ \lambda_{i+1} t^i \} \geq \min_{i \geq 0} \{ \lambda_i t^i \} \geq 0$$

Where the derivation of elements of $\mathbb{k}$ is zero so we do not descend to negative exponents. This shows $\dot{f} \in \mathcal{O}$.

The map $D: \mathfrak{R} \rightarrow \mathbb{K}$ given by writing elements of $\mathfrak{R}$ as their power series expansions and then taking their formal derivative, and then considering this as an element of $\mathbb{K} \leq \mathbb{k}((t))$, is therefore a derivation by Leibnitz's Rule. Moreover,

$$D[t] = \frac{\partial}{\partial t} t = 1$$

So is nontrivial. Hence $\Omega_{\mathbb{K}}[\mathbb{K}]$ is nonzero. ■